Early-Age Running Enhances Activity of Adult-Born Dentate Granule Neurons Following Learning in Rats

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By: Lynley, Tiana and Cass DV

Outline

Introduction

Hypothesis

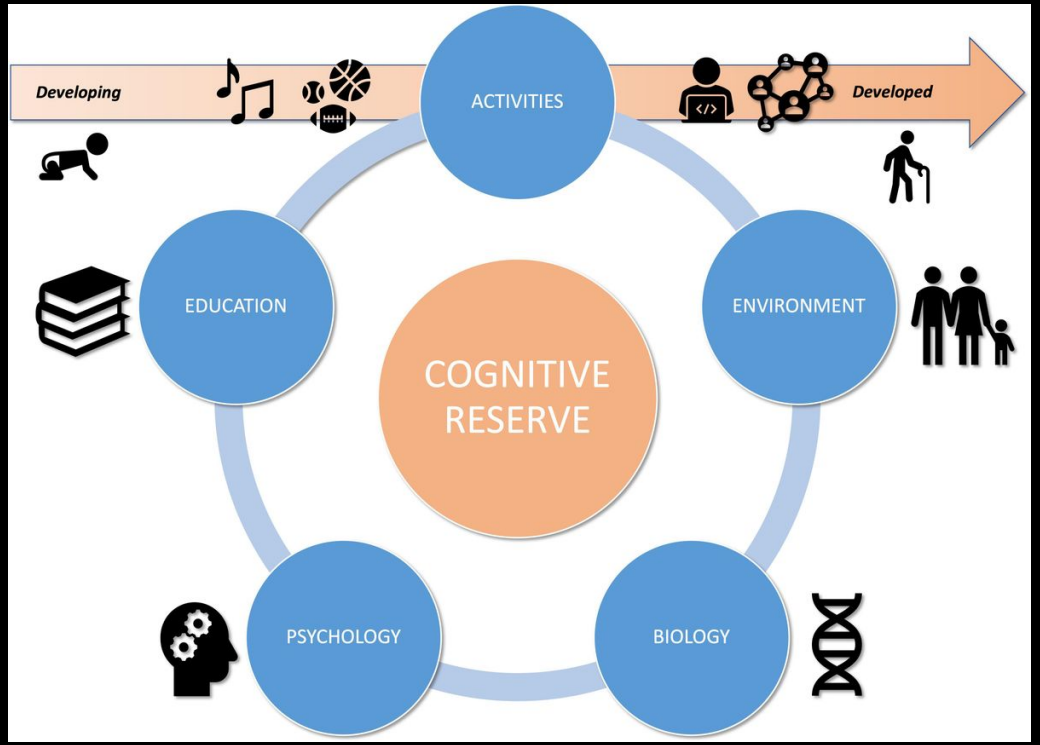
Experimental Approach

Results

Future directions

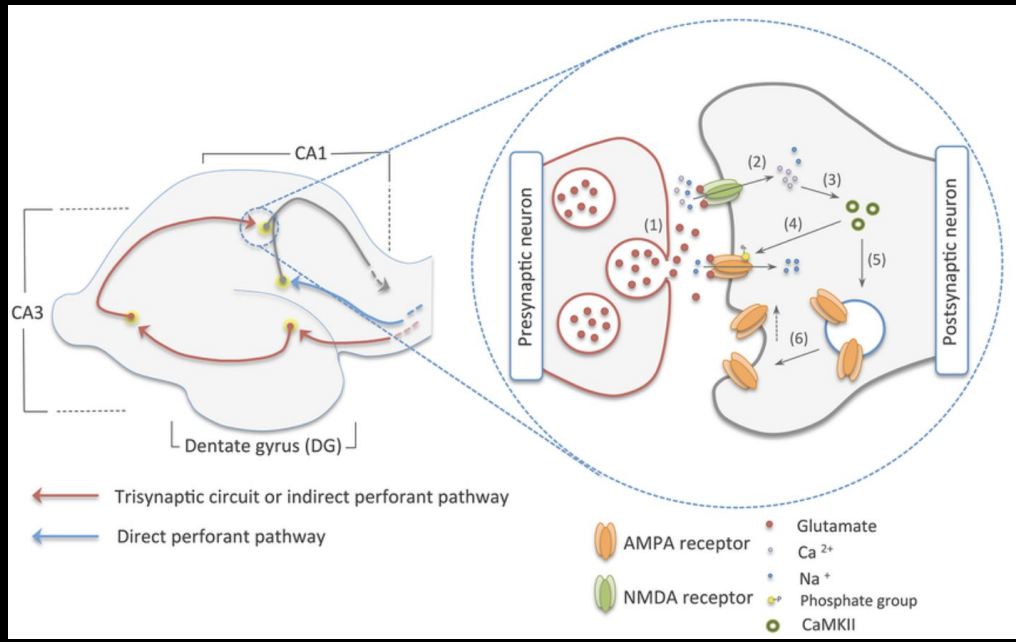
What is Cognitive Reserve?

- Brain's ability to use past experiences to resist decline in later life
- Lifestyle factors help build this reserve
- Individuals with ↑ cognitive reserve show better memory and problem solving skills



How does this relate to the brain's plasticity?

- Plasticity = brain's ability to adapt and form new connections in response to connections
- The hippocampus plays an important role in learning and memory and is one of the few regions that produces new neurons (adult neurogenesis)
- Exercise and enriched environments increase neurogenesis in the hippocampus
- Learning strengthens existing synapses in the hippocampus = Long term potentiation (LTP)

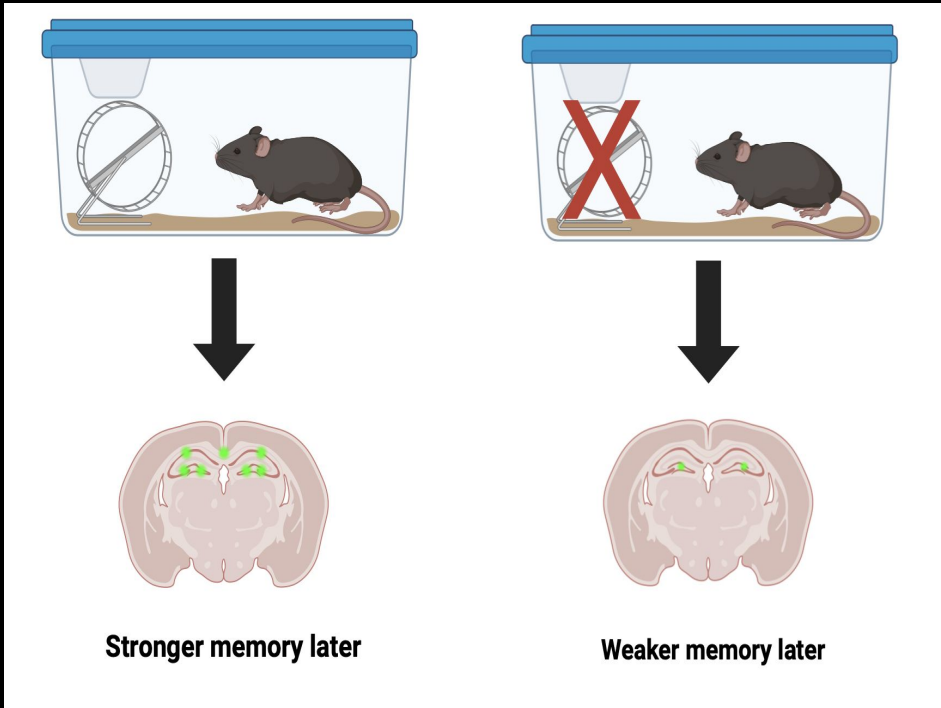


What we don’t know

- Does early enrichment lead to stronger activation or survival of new neurons later in life?
- Does this translate into better memory performance as adults?

Hypothesis

- Early running life ↑ the production of hippocampal neurons
- These neurons will be more active during memory task in adulthood
- This increased activity will support better memory and adaptability later in life

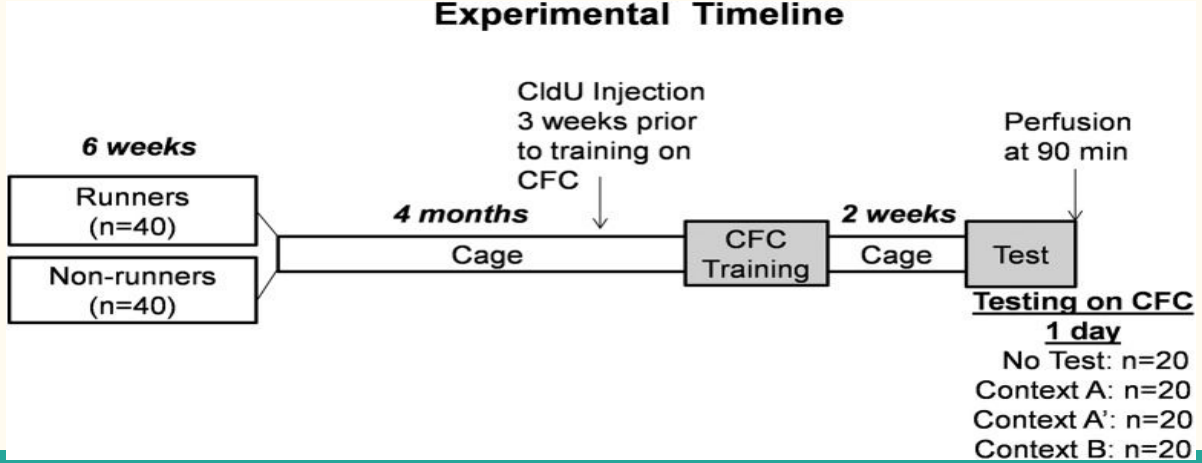


CldU: newly generated neurons

Fig. 1

Experimental Approach

- Subjects: one month old male rats
- Groups: Runners & Non-runners
- Brain region studied: Dentate Gyrus (hippocampus) = site of adult neurogenesis
- Approach: tested memory and activity in runners vs. non-runners using Contextual Fear Conditioning
    - Context A - same environment
    - Context A' - similar different environment
    - Context C - different environment

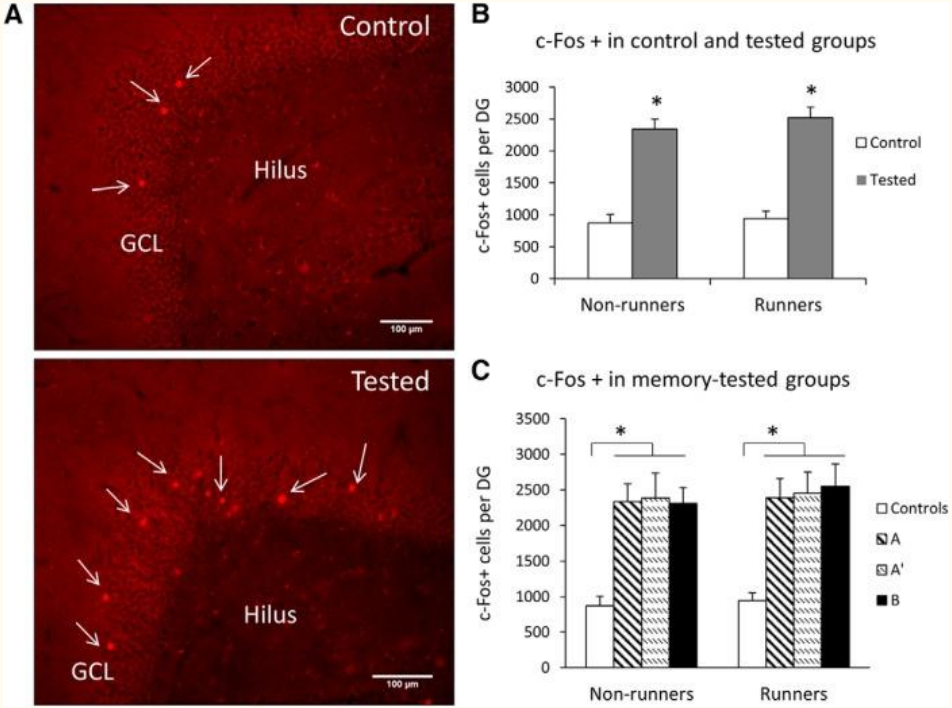


C-fos: marker of neuronal activity
CldU: newly generated neurons

Fig. 2
Results

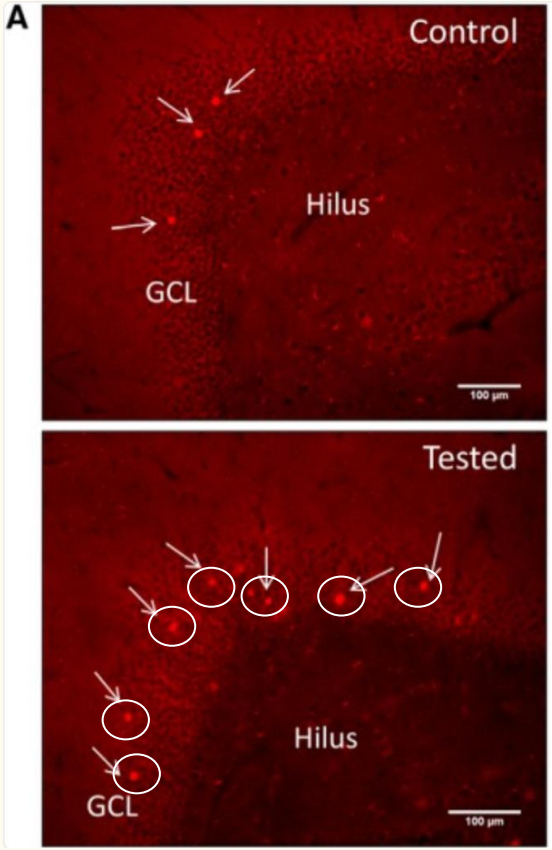
Results

Hippocampal neurons are activated during a contextual learning task



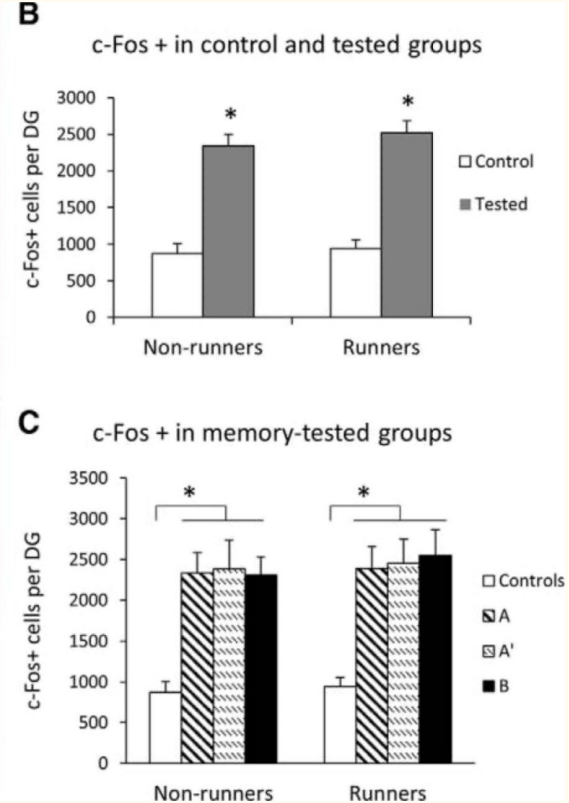
- The researchers wanted to measure the overall activity of hippocampal neurons in the dentate gyrus.
- For this reason, the researchers stained hippocampus neurons with the protein c-Fos to identify cell survival and activity of newly formed cells

Hippocampal neurons are activated during a contextual learning task



- Fluorescent microscopic images of c-Fos cells in the dentate gyrus of control and tested rats
- White circles: c-Fos-labeled cells in the GCL of the DG
- Tested group ↑ c-Fos positive cells per DG than controls

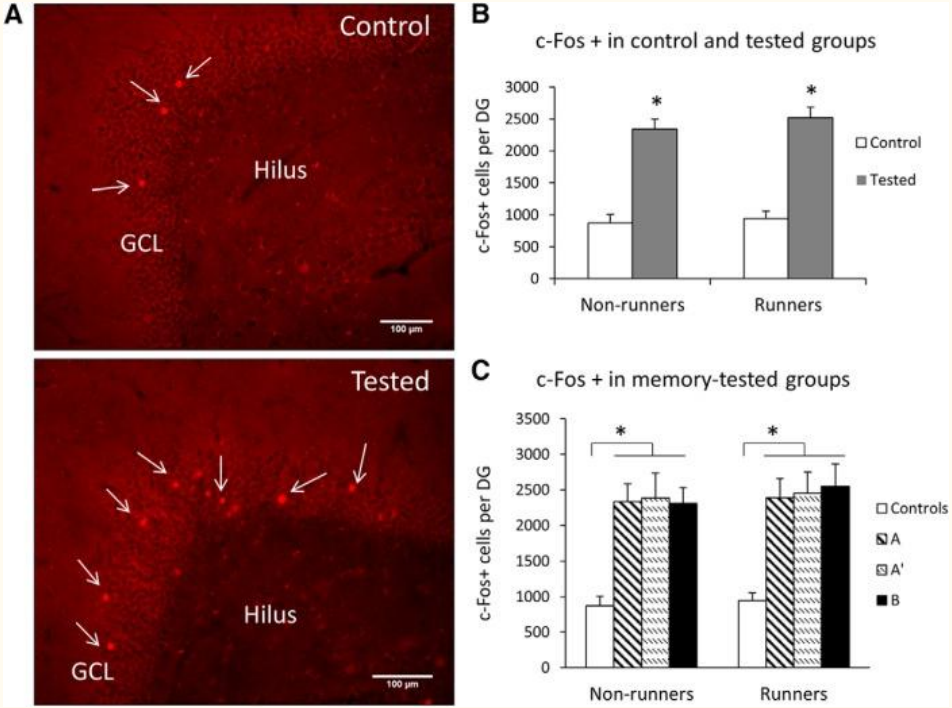
Hippocampal neurons are activated during a contextual learning task



- Quantifications of c-Fos positive cells per DG in control and tested groups and memory tested groups
- Tested groups ↑ c-Fos expression than controls
- Both Runners and Non-runners ↑ c-Fos positive cells compared to controls

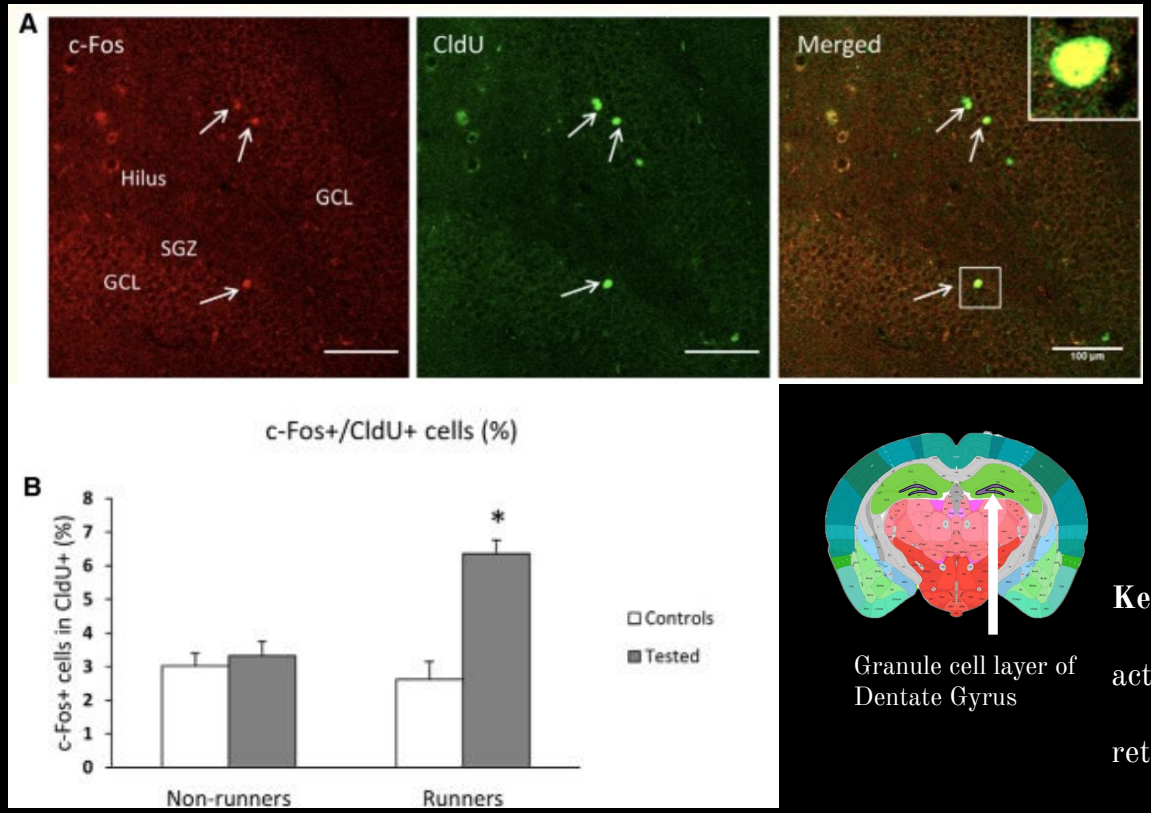
A: familiar environment; A': similar environment; B: a completely different environment

Hippocampal neurons are activated during a contextual learning task



- Key Takeaway:
  - Learning increases neuronal activation
  - This occurs in both runners and non-runners
  - Early-life running enhances how active the adult-born neurons get after we learn something.

Early-Age Running increases neuronal activity during adult memory tasks

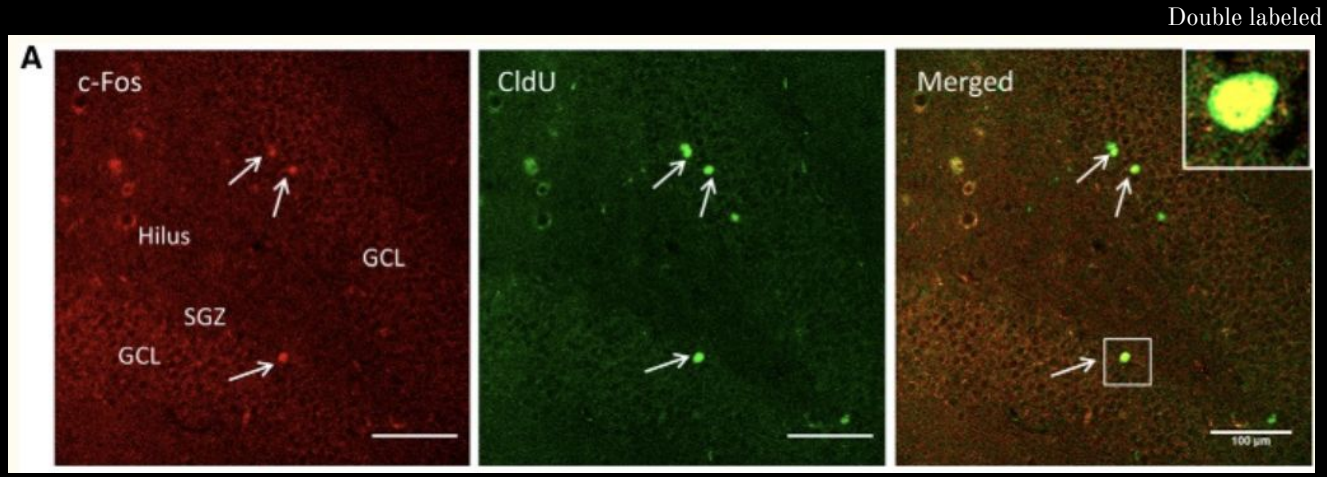


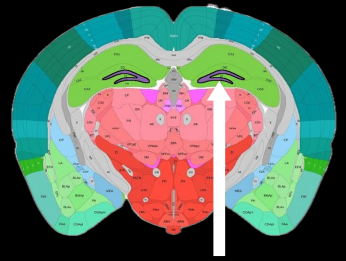
Granule cell layer of Dentate Gyrus

- c-Fos: marks neurons that were active during memory retrieval
- CldU: labeled neurons which were 5 weeks old at time of testing
- Measured % of CldU positive cells that were also c-Fos positive
- Runners had a higher % of CldU+/c-Fos+ cells

Key takeaway: Early-Age running increases the later activation of adult born neurons during memory retrieval.

Early-Age Running increases neuronal activity during adult memory tasks



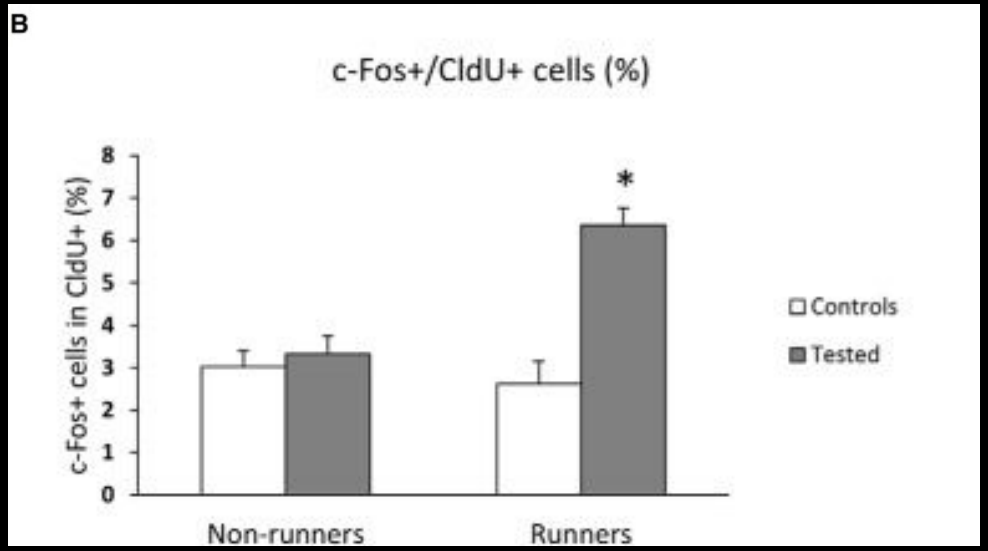


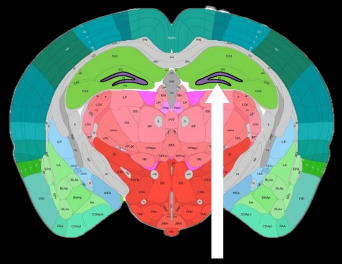
Granule cell layer of Dentate Gyrus

- c-Fos: neurons active during memory retrieval
- CldU: 5 week old adult born neurons
- **Merged**: overlap of both labels → active, adult-born neurons

- **Key takeaway:** Early-Age running increases the later activation of adult born neurons during memory retrieval.

Early-Age Running increases neuronal activity during adult memory tasks



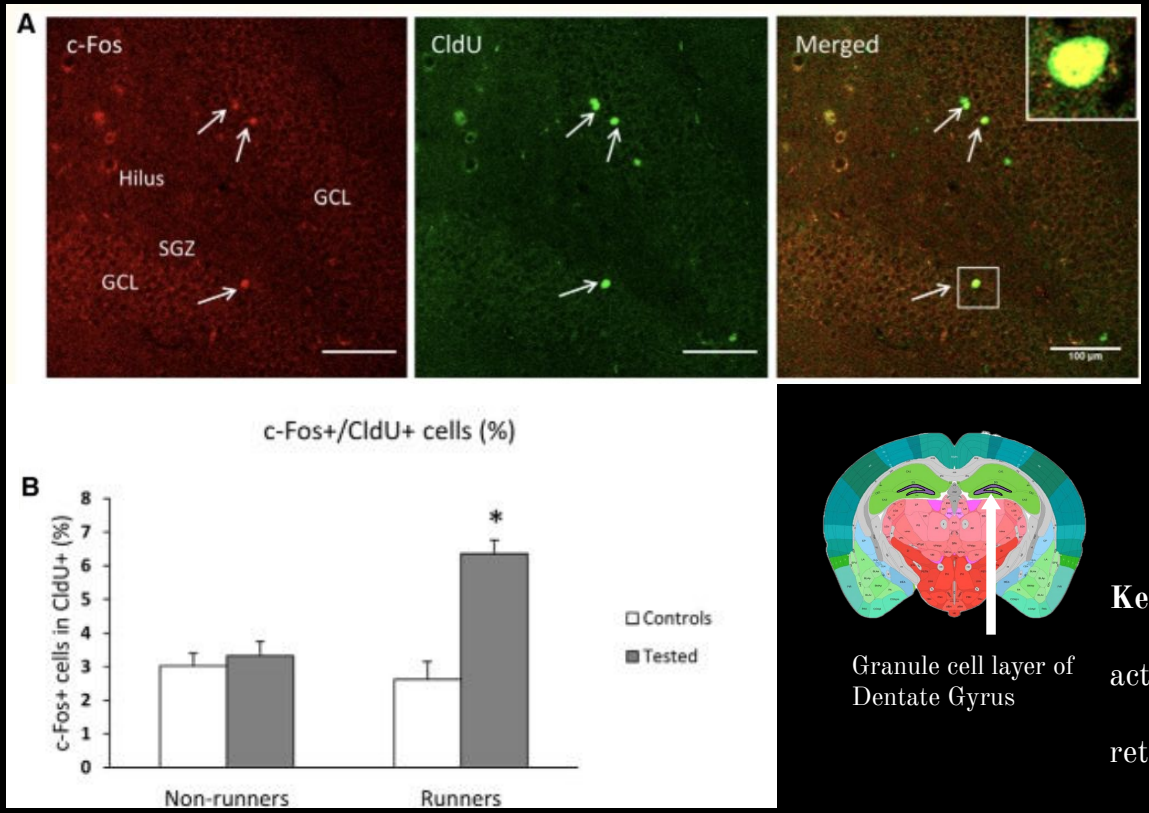


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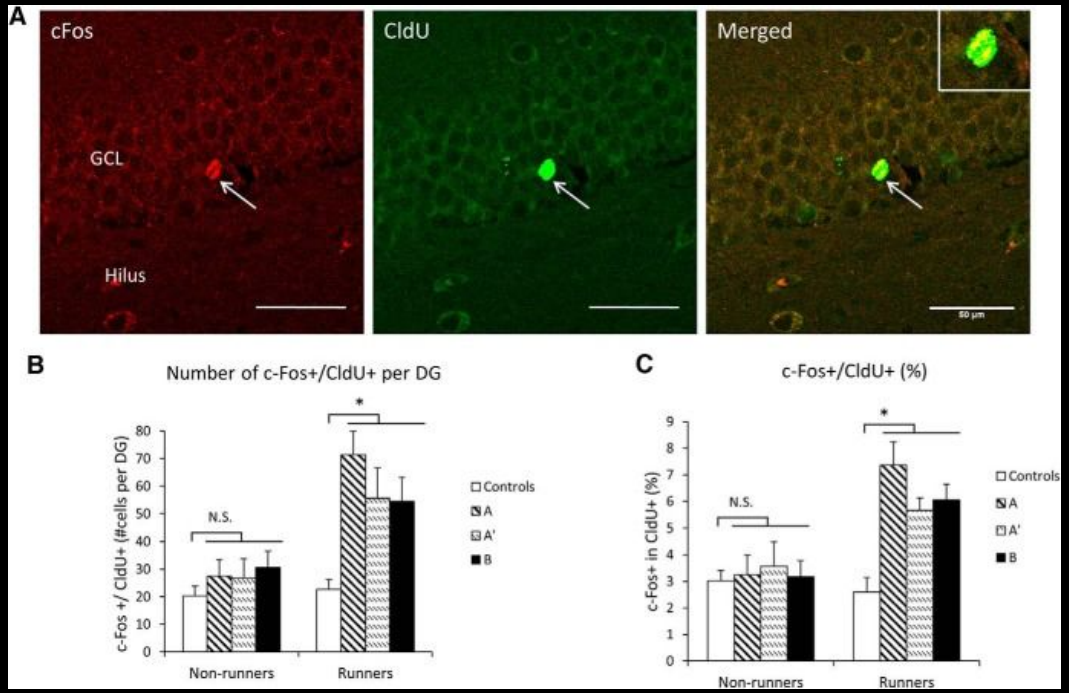
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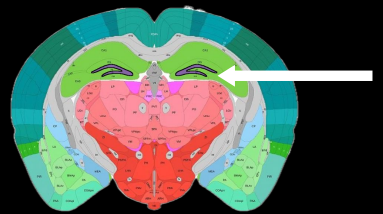
Fig. 4

c-Fos activity in adult-born neurons is enhanced by running regardless of task environment similarity



- c-Fos: marks neurons that were active during memory retrieval
- CldU: 5 week old adult born neurons
- A: Same testing environment
- A': Similar testing environment
- B: Novel testing environment

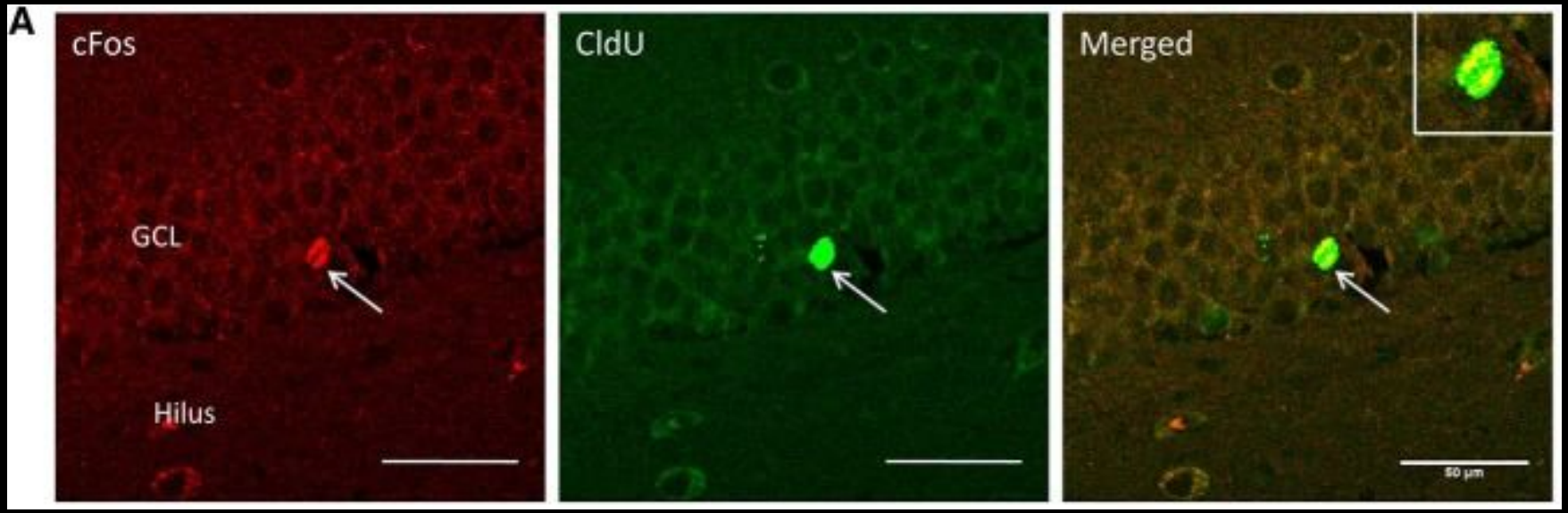
Key Takeaway: Engaging in running prior to reaching adulthood significantly increases active of adult born neurons regardless of testing environment



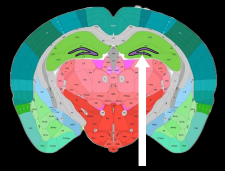
Granule cell layer of Dentate Gyrus

c-Fos activity in adult-born neurons is enhanced by running regardless of task environment similarity

Double labeled

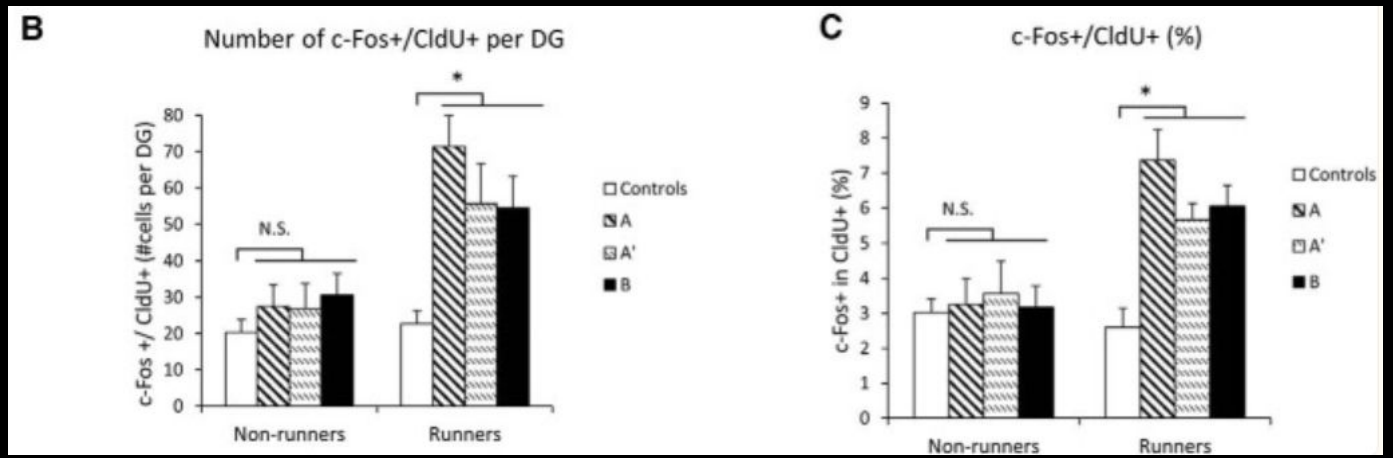


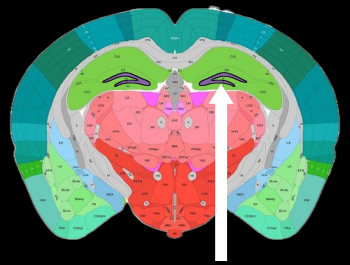
- c-Fos: marks neurons that were active during memory retrieval
- CldU: 5 week old adult born neurons
- Adult-born neurons were active after learning took place



Granule cell layer of Dentate Gyrus

c-Fos activity in adult-born neurons is enhanced by running regardless of task environment similarity





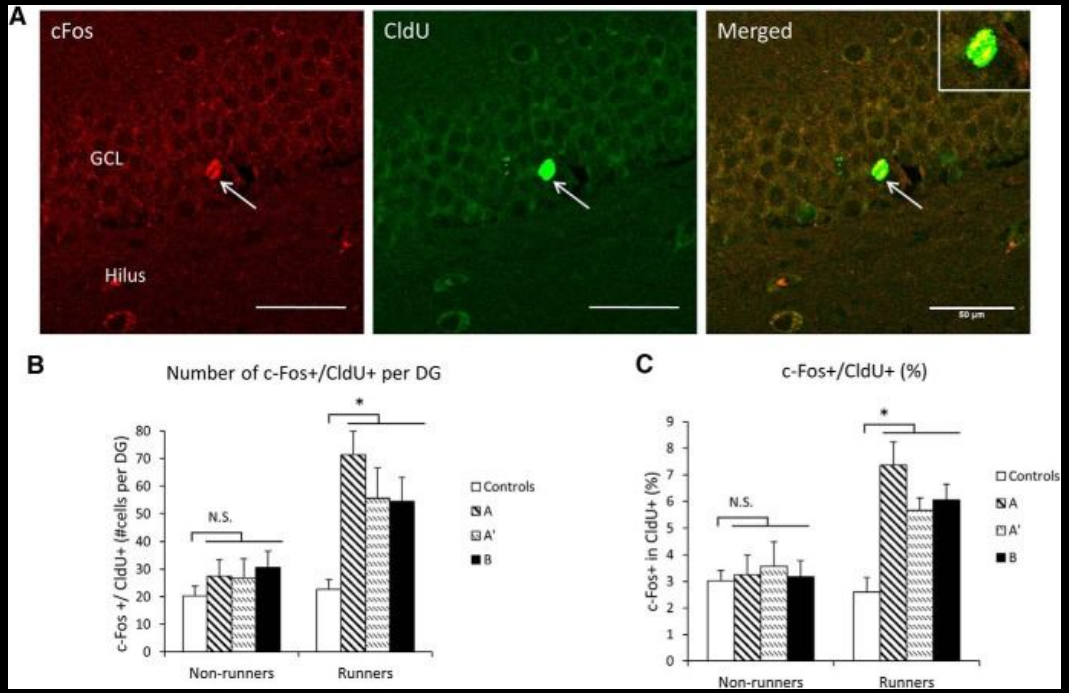
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- A: Same testing environment
- A': Similar testing environment
- B: Novel testing environment

- Engaging in running prior to reaching adulthood significantly increases ↑ c-Fos+/CldU+ regardless of testing environment

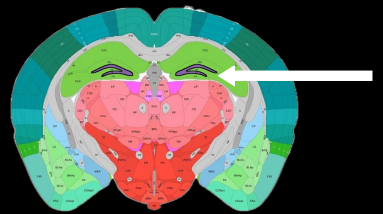
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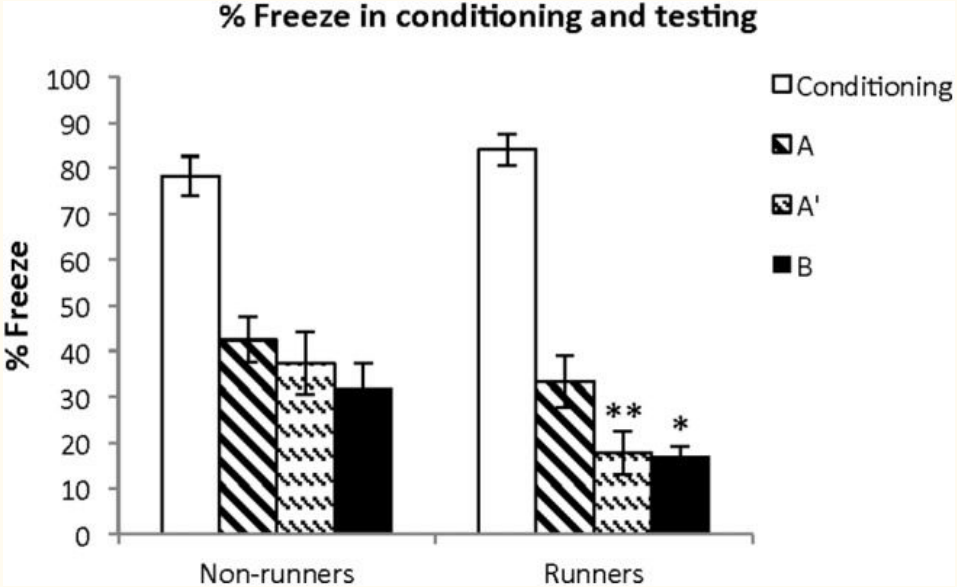
Key Takeaway: Engaging in running prior to reaching adulthood significantly increases active of adult born neurons regardless of testing environment



Granule cell layer of Dentate Gyrus

Fig. 5
(summary)

Memory selectivity is enhanced by running

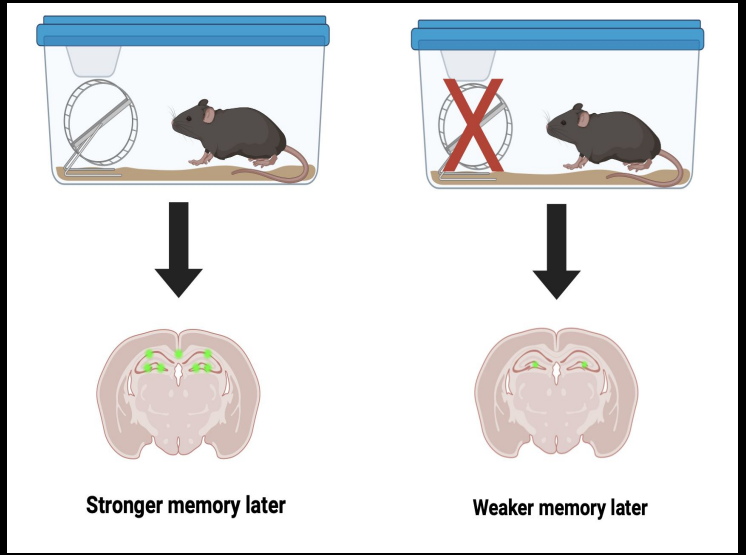


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- Figure 5 reveals that running during early life seems to boost memory specificity.
- This suggests that runners have a knack for better distinguishing between contexts.
- Plus, it lines up well with specific patterns of neuronal activation we see in the dentate gyrus.

Conclusion and Key Takeaways

Figure 1	Figure 2	Figure 3	Figure 4	Figure 5
Shows the overall experimental design used to test how early-age running influences adult neurogenesis and memory.	Performing the memory task increased neuronal activity in the dentate gyrus, regardless of whether the rats ran early in life.	Early-Age running increases the later activation of adult born neurons during memory retrieval.	c-Fos activity in adult-born neurons is enhanced by running.	Memory selectivity is enhanced by running



CldU: newly generated neurons

Future Directions

- Are there specific “critical periods” in development that exercise has the most effect on during neurogenesis and learning?
- How might these results be applied to humans?

Further studies:

Differential effects of adolescent and adult-initiated exercise on cognition and hippocampal neurogenesis

James D O'Leary 1, Alan E Hoban 1, Ashley Murphy 1, Olivia F O'Leary 1 2, John F Cryan 1 2, Yvonne M Nolan 1 2

Affiliations Expand, PMID: 30844139, DOI: 10.1002/hipo.23032